

Project BUDDY: An Emotionally Intelligent Robotic Companion with AI-Driven Security.

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The Problem

- Many individuals in modern urban environments, particularly the elderly or those living alone, experience loneliness and require companionship.
- Traditional home security systems function only as passive surveillance devices without interaction, mobility, or emotional intelligence.

Users and Stakeholders

- Smart home users
- Elderly care facilities
- Individuals living alone
- Remote workers
- Home security companies

Impact on Users and Society

- Reduces loneliness and emotional stress
- Improves safety monitoring in homes
- Removes the need for users to constantly monitor security cameras

Existing Solutions & Limitations

- Robotic pets like Sony Aibo are very expensive
- Voice assistants rely on cloud processing
- Cloud systems create privacy risks and response delays

Industry Comparison

Feature	Sony Aibo	Boston Dynamics Spot	Unitree Go1	Buddy Robot
Face Recognition	Yes	Limited	Limited	Yes
Stranger Detection	No	Yes	Limited	Yes
Emotion Detection	No	No	No	Yes
Companion Interaction	Yes	No	No	Yes
Emotion Monitoring	No	No	No	Yes

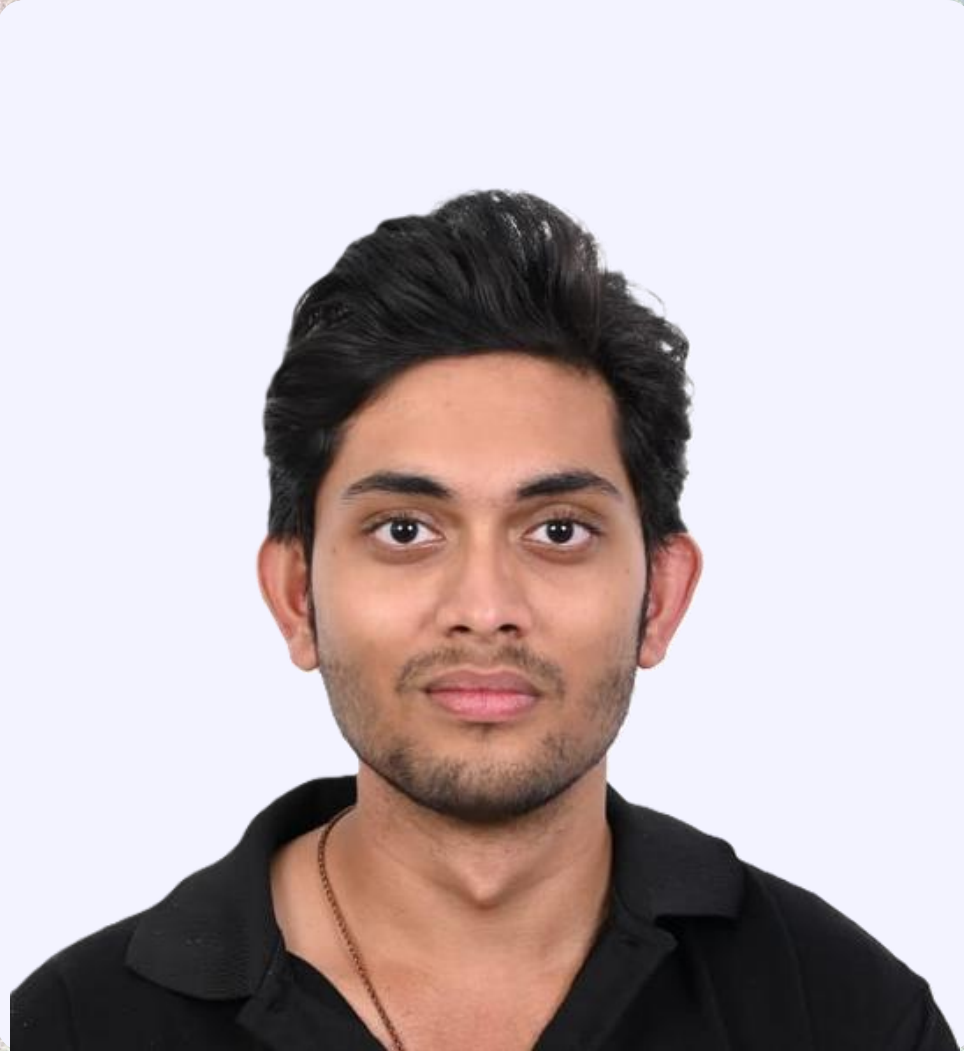
Proposed Solution

Buddy an affordable, emotionally intelligent robotic dog that processes all data locally via Edge AI to ensure privacy. It integrates intelligent indoor navigation, facial/emotion recognition, and adaptive voice interactions.

Methodology

- Problem Analysis
- System Design
- Data Collection & Model Training
- AI Model Development
- Hardware Integration
- System Testing & Evaluation

The Team



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Offline Voice Command Recognition

Enable Buddy robot to understand and execute commands without internet connectivity

Offline Voice Command Recognition

Novelty

- ML-based intent parser
- Optimized speech recognition for Sri Lankan English accents
- Emotion-adaptive speech synthesis

Problem Definition

Most existing voice assistants depend on cloud processing which causes:

- Privacy risks
- Slow responses
- Internet dependency

Research Objective

Develop a **fully offline voice interaction system** using:

- Wake-word detection
- Local speech recognition
- Intent classification

Offline Voice Command Recognition

Application of Domain Knowledge

- Natural Language Processing (NLP)
- Speech Processing
- Edge Artificial Intelligence

Technologies Used

- **Porcupine** – Wake word detection
- **Vosk** – Offline Speech Recognition
- **TF-IDF + Logistic Regression / TinyBERT** – Intent Classification
- **pyttsx3** – Text to Speech

Data Availability & Ethics

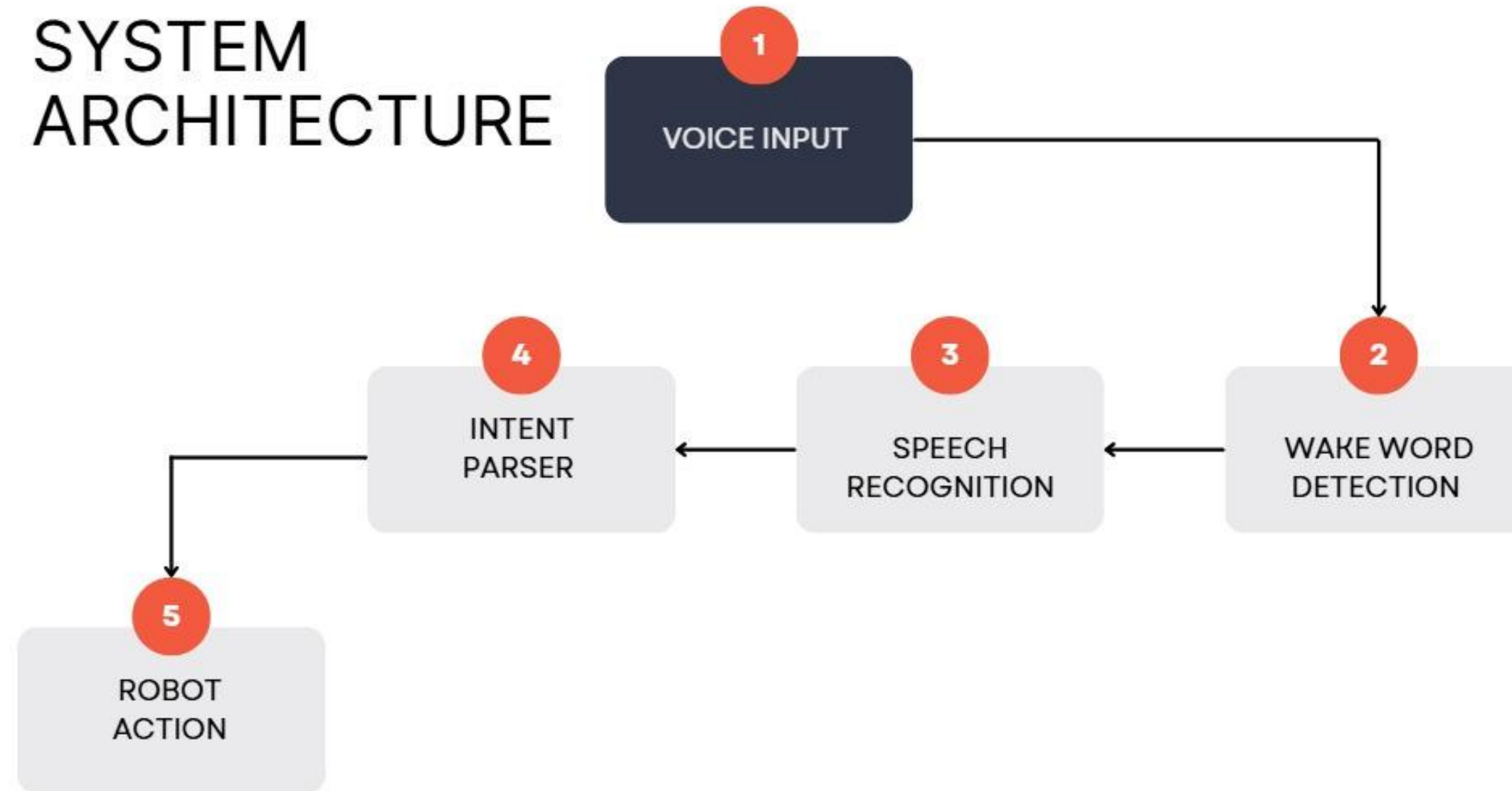
- All voice processing happens locally on device
- No user audio stored in cloud
- Dataset built with Sri Lankan English accents with user consent

Validation and Evaluation

- Wake word detection accuracy
- Word Error Rate (WER)
- Command execution success rate
- Response latency (< 2 seconds)

Offline Voice Command Recognition

SYSTEM ARCHITECTURE



Dependencies

- Raspberry Pi 4 Model B
- Microphone module
- Communication with:
 - Face recognition module
 - Navigation system



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Autonomous Navigation and Obstacle Avoidance

Allows the robot to move inside the house, follow users and avoid obstacles

Autonomous Navigation and Obstacle Avoidance

Novelty

Hybrid Control System via Priority-Based Subsumption Architecture.

Problem Definition

Traditional navigation systems:

- Have jerky movements
- Cannot handle dynamic indoor environments
- Are expensive and proprietary

Research Objective

Develop a navigation system capable of:

- Human tracking
- Field of View based perception
- Stable movement control

Autonomous Navigation and Obstacle Avoidance

Application of Domain Knowledge

- Robotics
- Control Systems
- Subsumption Architecture

Technologies Used

- ESP32
- Raspberry Pi 4
- ROS
- Ultrasonic/LiDAR Sensors.

Safety Considerations

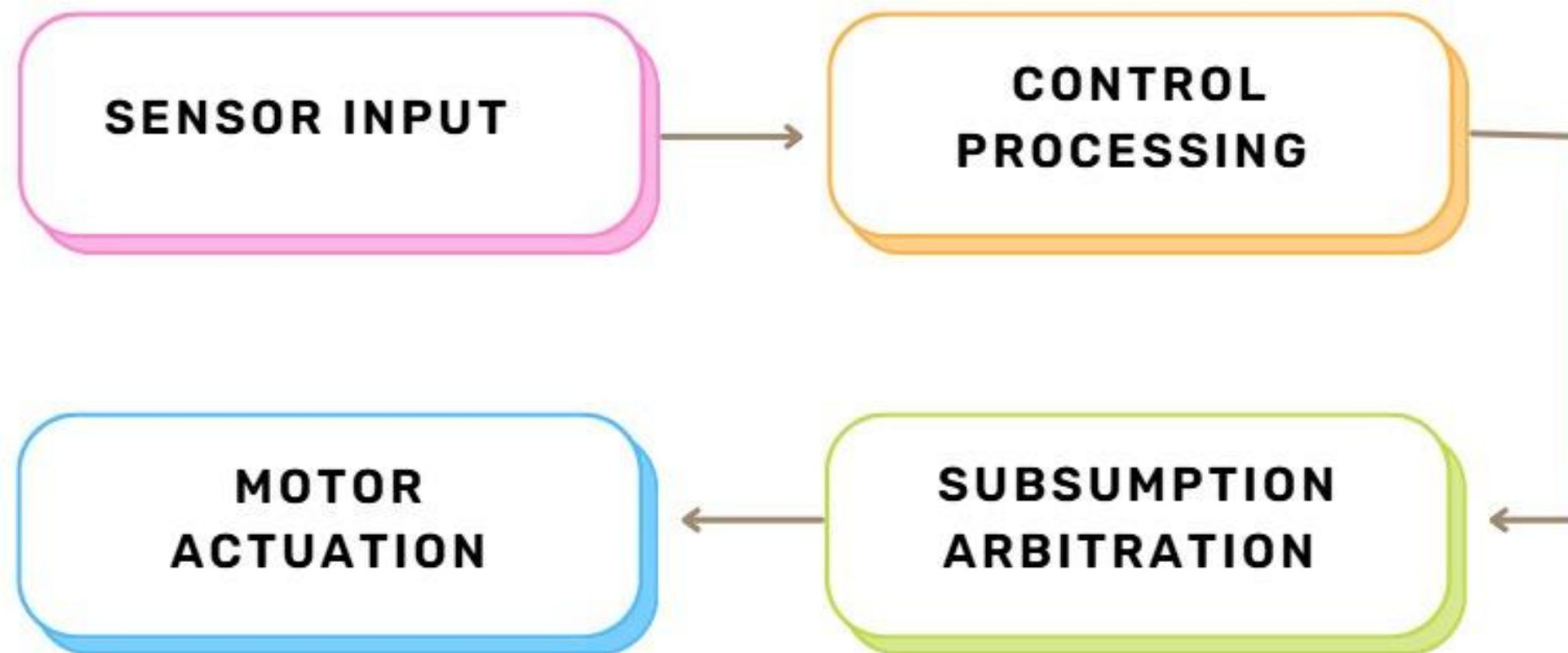
- Collision prevention
- User safety prioritization
- Hardware-level reflexive override.

Validation and Evaluation

- Indoor navigation simulations
- Human-following tests
- Obstacle avoidance reliability tests

Autonomous Navigation and Obstacle Avoidance

SYSTEM ARCHITECTURE

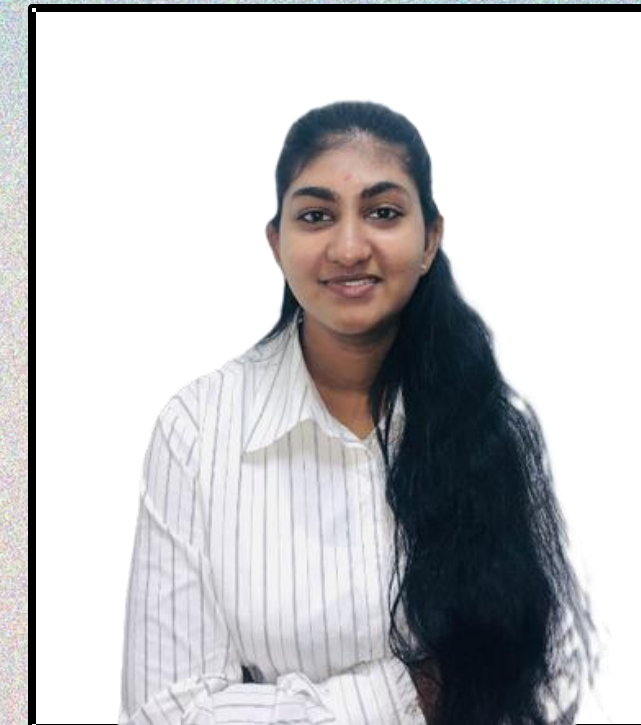


Dependencies

Raspberry Pi Camera
Ultrasonic Sensors
LiDAR Sensors

Integration

Works with:
Voice command module
Face recognition system
Example Commands:
"Follow me"
"Patrol the house"



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Face Recognition and Emotion Detection Subsystem.

Identifies users and strangers detects emotional state using AI and understand humans emotions and adapts its behaviour to it

Face Recognition and Emotion Detection Subsystem

Novelty

- Combines security and emotional monitoring
- Motivational voice responses when negative emotions detected
- Fully offline processing

Problem Definition

Traditional cameras:

- Require manual monitoring
- Send biometric data to cloud servers

Research Objective

Develop a system capable of:

- Face detection
- User identification
- Emotion recognition

Face Recognition and Emotion Detection Subsystem

Application of Domain Knowledge

- Computer Vision
- Deep Learning
- Affective Computing

Technologies Used

- MTCNN(Multi-task Cascaded Convolutional Neural Network) – Face Detection
- FaceNet – Face Embedding
- DeepFace – Emotion Detection
- TensorFlow
- OpenCV

Evaluation Metrics

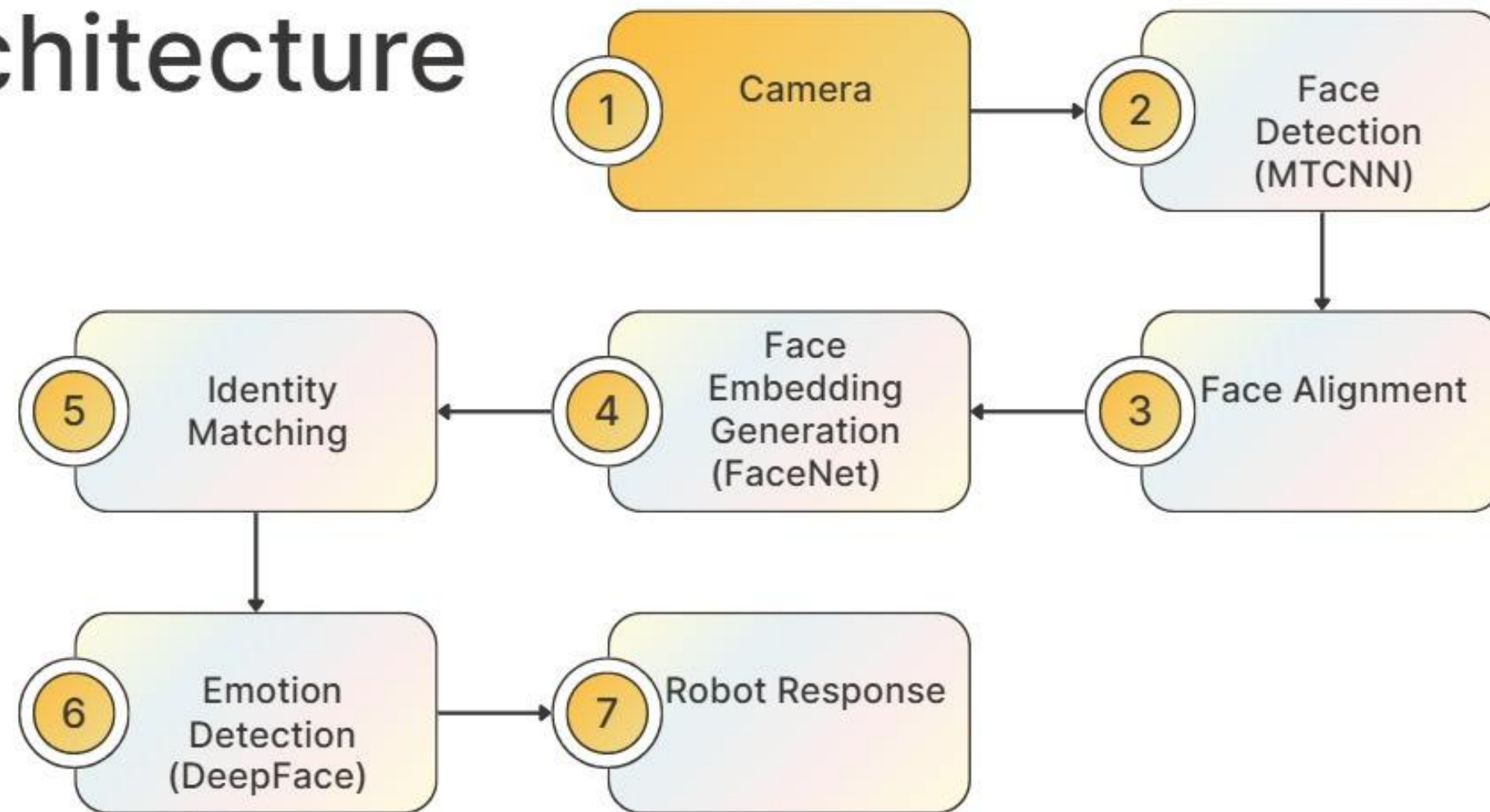
- Face recognition accuracy
- FAR and FRR
- Emotion detection accuracy
- Real-time processing speed

Data Ethics

- Uses FER2013 dataset
- Stores only face embeddings (not images)

Face Recognition and Emotion Detection Subsystem

System Architecture



Dependencies

- Raspberry Pi 4
- Pi Camera

Integration

Communicates with voice module to trigger responses based on emotion.

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Commercial Potential

Smart home robotics
Elderly care systems
Personal home security

Estimated Investment

Prototype Cost ≈ **LKR 60,000**

Major Components:
Raspberry Pi 4 – 18,000 LKR
Servo Motors – 20,000 LKR
Arduino Uno – 5,000 LKR

Target Market

- Smart home users
- Elderly individuals
- Small security companies

Value Proposition

Affordable robotic companion combining:

- Security monitoring
- Emotional interaction
- Offline AI privacy protection

Revenue Model

- Robot hardware sales
- Software upgrades
- Mobile app subscription

WBS & Gantt Chart



Work Breakdown Structure

1. Research and system design
- 2.Dataset collection
- 3.AI model development
- 4.Hardware integration
- 5.Sensor deployment
- 6.Testing and optimization
- 7.Final evaluation

**Thank
You!**